

Introduction to Machine Learning

Machine learning is a subset of artificial intelligence that focuses on building systems that learn from data. Instead of being explicitly programmed, these systems improve through experience.

There are three main types of machine learning:

1. Supervised Learning - Uses labeled data to train models
2. Unsupervised Learning - Finds patterns in unlabeled data
3. Reinforcement Learning - Learns through trial and error

Common algorithms include linear regression, decision trees, neural networks, and support vector machines.

Deep Learning and Neural Networks

Deep learning uses artificial neural networks with multiple layers to progressively extract higher-level features from raw input.

Convolutional Neural Networks (CNNs) are widely used for image recognition tasks. They use convolutional layers to detect features like edges, textures, and shapes in images.

Recurrent Neural Networks (RNNs) are designed for sequential data such as text and time series. LSTM and GRU variants solve the vanishing gradient problem in long sequences.